

Supraventricular tachycardia

Supraventricular tachycardia (SVT) is defined as a narrow complex tachycardia. It is the most common type of tachyarrhythmia in children.

Causes of SVT:

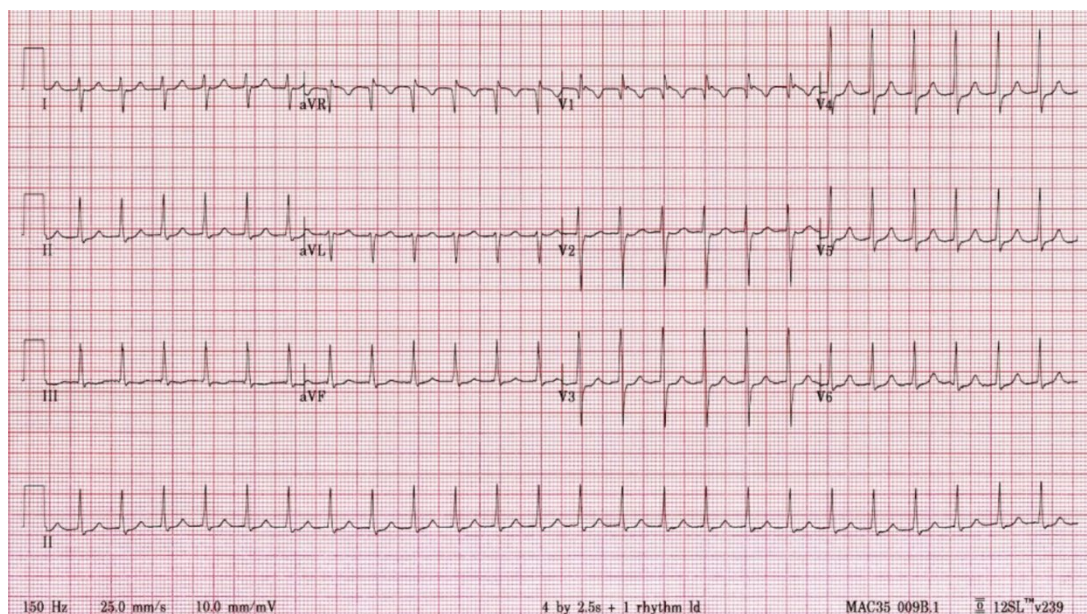
- 1) Nodal (AV junction) tachycardia (AVNRT)
 - 2) AV reentrant tachycardia (AVRT)
 - 3) Atrial tachycardia (Focal atrial tachycardia, Multifocal atrial tachycardia)
 - 4) Others: Sinus tachycardia
- No heart disease is found in about half the patients.
 - Wolff parkinson white (WPW) pre excitation occurs in 10-20% of cases, which is only evident only after conversion to sinus rhythm
 - Some congenital heart defects (Ebstein's anomaly, single ventricle, congenitally corrected transposition of the great arteries) are more susceptible to this arrhythmia

Clinical features suspicious of SVT:

- History and physical examination incompatible with sinus tachycardia
- Neonates or infants with persistent crying, irritability, tachypnoea, poor feeding, pallor
- Older children may complain of palpitations, chest pain, shortness of breath, lightheadedness, fatigue
- Sudden onset tachycardia with no other cause
- General guide >180 beats per minute in children and >220 beats per minute in infants

ECG findings:

- Narrow complex tachycardia
- Absent p waves
- No beat to beat variability
- Remember to look for delta wave after cardioversion to identify patients with WPW syndrome

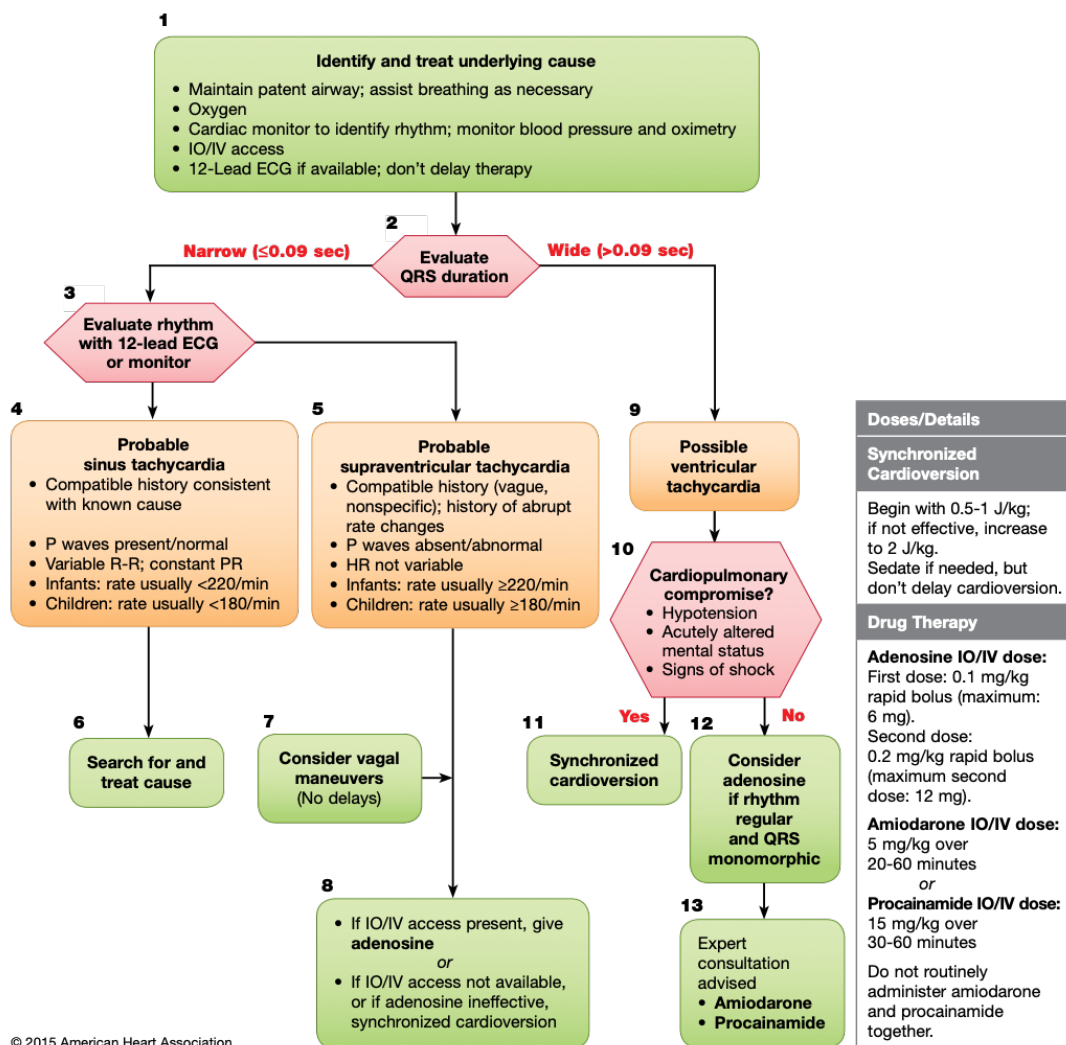


12 lead ECG with SVT

Approach to a patient with SVT:

- 1) Place on continuous cardiac monitoring
- 2) Assess for presence and strength of peripheral pulses (clinical evidence of low cardiac output like dizziness)
- 3) Take a blood pressure to differentiate between stable and unstable SVT
- 4) Give supplemental oxygen via face mask or non-rebreather mask
- 5) Attach 12-lead ECG
- 6) Look for correctable factors: fever, electrolyte abnormalities, dehydration, illicit drug use especially those that acts on the sympathetic nervous system
- 7) Obtain IV access, preferably over the right upper limb at the cubita fossa where it has the shortest venous route towards the heart. If unable to obtain venous access, consider intraosseous insertion
- 8) Prior to administering pharmacological agents or cardioversion, allow 12-lead ECG to run continuously

Pediatric Tachycardia With a Pulse and Poor Perfusion Algorithm

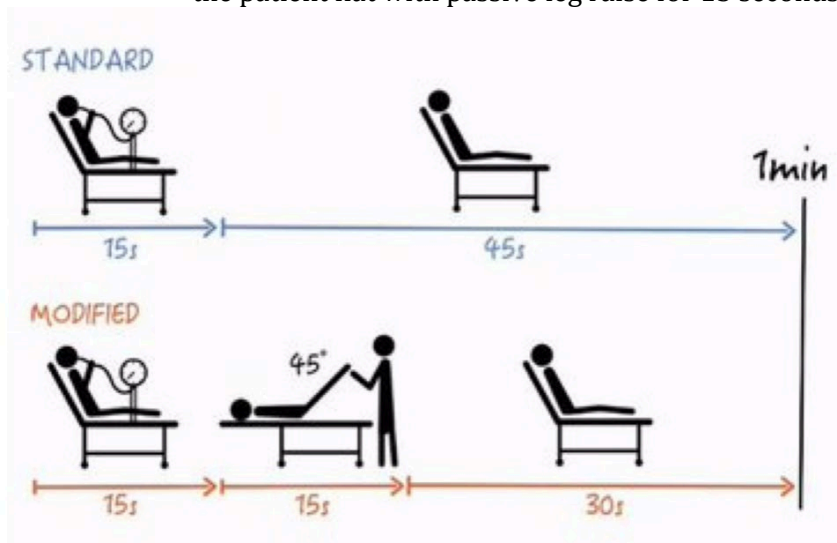


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Management:

- 1) Vagal maneuvers
 - Do not delay administration of Adenosine due to vagal manoeuvres
 - Place a bag with ice water over the upper half of the infant's face (without obstructing the airway)

- Modified Valsava (REVERT Trial): In an older child, place the child in semi recumbent position 45° on the trolley and ask them to blow into a syringe with the plunger in place for 15 seconds, then immediately lie the patient flat with passive leg raise for 15 seconds

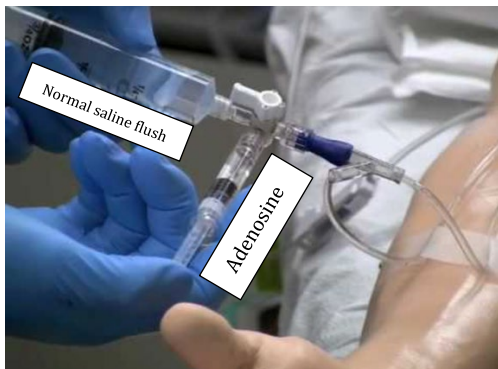


Postural modification to the standard Valsava manoeuvre for emergency treatment of supraventricular tachycardias (REVERT): a randomized controlled trial. Andrew Appelboam et al.

- No carotid massage in children

2) Adenosine

- Blocks conduction through the AV node temporarily
- Must cause AV block with sinus pause
- Half-life short at 10 seconds
- Administer via rapid IV bolus with a 3-way tap connected to Adenosine and normal saline flush (minimum 10ml syringe)
- 1st dose: IV Adenosine 0.1mg/kg (maximum 6mg)
- 2nd dose: IV Adenosine 0.2mg/kg (maximum 12mg)



3) Amiodarone

- CALL CARDIOLOGIST! If you are considering given Amiodarone
- Preferred due to IV formulation
- Mechanism of action: inhibits alpha- and beta- adrenergic receptors, producing vasodilation and AV nodal suppression, slowing conduction through AV node. Inhibits the outward potassium current so it prolongs the QT duration. Inhibits sodium channels, which slows conduction in the ventricles and prolongs QRS duration

- Rare but significant acute side effects include bradycardia, hypotension, polymorphic ventricular tachycardia (VT)
- Long half-life, once administered, effect will remain
- Dose: usually less than ventricular fibrillation dosing
 - ◆ VF dosing: IV Amiodarone 5mg/kg over 20-60 minutes
 - ◆ To discuss with Cardiologist regarding dosing, usually 1mg/kg over 20-60 minutes
- Rapid administration may cause vasodilation and hypotension
- Dilute in D5%
- For peripheral infusion through IV plug: concentration $\leq 2\text{mg/mL}$
- Monitor blood pressure frequently during administration

4) Synchronised cardioversion

- Synchronised cardioversion allows for shock delivery to coincide with the R wave of the patient's QRS complex. The goal is to prevent ventricular fibrillation (VF) that could result from delivery of the shock during the vulnerable period of the T wave
- Indications: SVT not responding to pharmacological therapy, unstable patients with hemodynamic compromise such as unconsciousness or shock
- Remember to give analgesia prior to cardioversion such as IV Fentanyl 1-2mcg/kg
- 0.5J/kg followed by 1J/kg, followed by 2J/kg (maximum dose 50-100J)
- Round up to the next available whole number if calculated dose is not available

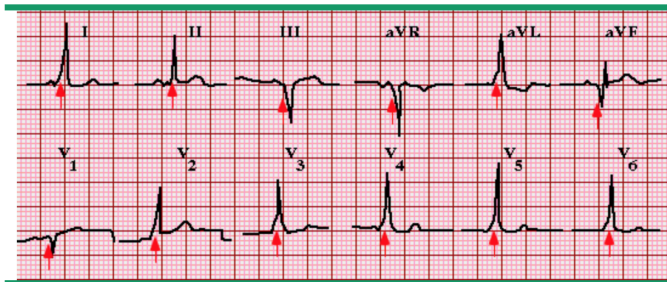
Post SVT-conversion management

- Obtain a 12-lead ECG in sinus rhythm
- Look for delta waves which may suggest underlying cause of WPW
 - ECG features of WPW: delta wave, wide QRS complex (>0.12 seconds), short PR interval (<0.12 seconds)



WPW, with delta wave (arrow)

12-lead electrocardiogram (ECG) showing the Wolff-Parkinson-White pattern



The two main electrocardiographic features of Wolff-Parkinson-White (WPW) pattern include a short PR interval (<0.12 seconds) and a delta wave (arrows). The QRS complex is wide (>0.12 seconds) and represents a fusion beat; the initial portion (delta wave) results from rapid ventricular activation via the accessory pathway (preexcitation), while the termination of ventricular activation is via the normal conduction system, leading to a fairly normal terminal portion of the QRS.

Disposition:

Criteria for discharge

- Stable SVT, reverts with non-pharmacological measures or Adenosine
- >6 years old
- Normal rhythm and hemodynamics after 2 hours of observation from time of cardioversion
- Compliant patients who will likely return for follow up
- Parents/ child confident of identifying recurrence of SVT
- Prior to discharge:
 - Patients with known SVT, contact cardiologist regarding need for adjustment of medications, with the possibility of discharge if patient/ parents are confident
 - First presentation of SVT: contact cardiologist regarding need for initiation of medications, with the possibility of discharge if patient/ parents are confident

Contraindications to discharge

- <6 years old
- Requiring drugs other than adenosine to revert
- Unstable SVT requiring synchronized cardioversion
- Hemodynamic instability
- Parents not confident
- Disposition to HD: patients who require synchronized cardioversion

References:

- 1) 2015 ACC/AHA/HRS Guideline for the Management of Adult Patients With Supraventricular Tachycardia. A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. Richard L. Page et al.
- 2) Postural modification to the standard Valsalva manoeuvre for emergency treatment of supraventricular tachycardias (REVERT): a randomised controlled trial. Andrew Appelboom, Adam Reuben, Clifford Mann, James Gagg, Paul Ewings, Andrew Barton, Trudie Lobban, Mark Dayer, Jane Vickery, Jonathan Bengier, on behalf of the REVERT trial collaborators.

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